

Human Classifier Simulation

Project: Transparent Book Recommender

1. Human Classifier Simulation

Teach the learning cycle

Primary target: Construct a classifier with a stated input format, selected rule and recorded output

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Case exposure check: this source example appears in ai-1/14, ai-2/10. Reuse a familiar case for recall or

modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Features and Distractors

Without opening the old lesson, explain one key idea from Features and Distractors. Give an example and a mistake that the idea helps you avoid.

Look for: Choose recorded properties that preserve distinctions needed for a stated task

Reference example, not the only acceptable wording: Use length and width measured before the decision. State the order as length in cm, then width in cm; the record is 12, 8. The target is fits or too large under the guide, and this card is fits because both equalities are included. Colour and card number are not needed for this rule; the final inspection sticker copies the answer and is unavailable before inspection.

Rows Columns and Data Types

Without opening the old lesson, explain one key idea from Rows Columns and Data Types. Give an example and a mistake that the idea helps you avoid.

Look for: Specify a table schema

Reference example, not the only acceptable wording: activity: nonempty text; minutes: integer from 0 to 60 inclusive.

Trace Tables

Without opening the old lesson, explain one key idea from Trace Tables. Give an example and a mistake that the idea helps you avoid.

Look for: Trace variable values before and after each loop iteration

Reference example, not the only acceptable wording: 2, 3 and 4; the stop value is excluded. The body runs three times.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Recommend a book only if genre is science AND pages $\leq$ 150. Classify science/150, science/151, mystery/90 and missing-genre/80. State a missing-data policy.

Concept to explain

Week 10: Human Classifier Simulation

Term · Meaning

Term: classifier

Meaning: A rule or model that assigns an input to a category.

Term: boundary

Meaning: A dividing condition between categories, actions or allowed behaviours.

Term: training example

Meaning: An input and any associated label used to fit a model.

Term: prediction

Meaning: An estimated category, value or outcome that must be distinguished from established fact.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Students construct a paper procedure around a fitted threshold, joining input checks, conditions, stopping and recorded output. Separate input validation, predictive correctness and subsequent human action. Tower1's range and whole-number requirement are classroom restrictions, not general requirements for classifiers.

Readiness answer

At 5 predict short; at 6 predict tall. Check that a height was supplied. A blank is missing information, not the number zero.

Input cards with 1, 5, 6, 9, 10 and 11 cm, a blank, 60 mm and 4.5 cm

Blank paper for numbered steps or branch diagrams and a record with input, version, outcome and reason

Separate Ribbon1 task cards; no accounts, devices or personal measurements required

End goal for the pictorial case

By the end, I can simulate a classifier with explicit requirements and missing-value behavior.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

1. Examples suggest: striped AND small → glim.
2. Learners infer a boundary from labeled cards.
3. New case: striped, small creature → predict glim.
4. Different inferred rules may match training yet disagree on a new case.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

Recommend if adventure AND pages<150; otherwise Other.

Trace A=(adventure,120), B=(adventure,180), C=(science,90).

Exact explanation to model

A is Recommend. B fails length; C fails genre. Show both conditions.

Explain the added worked example

Your task is to build a paper classifier that another learner can operate without guessing your instructions. It will predict short or tall for imaginary towers. Use the threshold of 6 centimetres selected after adding the height 5 short example in Week 13. Name this selected model Tower1.

Write an input contract: one height recorded as a whole number of centimetres from 1 through 10, including both endpoints. Require the unit cm to be shown. This is a classroom design choice about what the procedure accepts. It is not a claim that every accepted height appeared in fitting or that towers outside this range cannot exist.

A complete classifier needs instructions for checking the input, applying Tower1 when the input is accepted, and recording the outcome. It must also say what happens when the input fails a check. The threshold alone does not provide these surrounding instructions.

For every record, first check that a height is present and its stated unit is cm. Then check that it is a whole number from 1 through 10. If any check fails, record no prediction and the reason, then stop processing that record. Do not continue into the threshold comparison.

If all checks pass, compare the height with 6. At least 6 gives the prediction tall; otherwise give short. Record the original input, model version Tower1 and the prediction. For a rejected record, record the original input, Tower1, no prediction and a reason. Tower1 identifies the intended model even when its threshold was not used.

Key terms

Term · Meaning

Term: Classifier

Meaning: A procedure that assigns a predicted category to an input

Term: Input contract

Meaning: The format and conditions an input must satisfy for a procedure to accept it

Term: Validation

Meaning: Checking whether an input satisfies the stated contract

Term: No prediction

Meaning: An explicit outcome when the classifier does not assign a category

Term: Model version

Meaning: A name that identifies the particular model used for a prediction

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

1 cm · Accepted at lower endpoint · Predict short

1 cm: 9 cm

Accepted at lower endpoint: Accepted

Predict short: Predict tall

1 cm: 10 cm

Accepted at lower endpoint: Accepted at upper endpoint

Predict short: Predict tall

1 cm: 0 cm

Accepted at lower endpoint: Outside 1 through 10

Predict short: No prediction

1 cm: 60 mm

Accepted at lower endpoint: Unit is not cm

Predict short: No prediction

1 cm: 4.5 cm

Accepted at lower endpoint: Not a whole number

Predict short: No prediction

A valid 9 cm input receives a tall prediction, but the input check does not establish that the prediction matches its comparison label. Validation checks the contract. Evaluation checks predictions against appropriate evidence. Keep these questions separate.

Sixty millimetres equals six centimetres, but this version accepts cm records only. It does not contain a conversion step. Ask for a correctly recorded cm measurement or deliberately design and check a conversion step in a new version. Do not silently treat the digits 60 as centimetres. A missing height must not become an invented zero either.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Human Classifier Simulation

A fictional librarian wants illustrated mystery books with at most 100 pages.

The facts we will use

Written rule: choose only genre mystery AND illustrated True AND pages $\leq$ 100.

Rows J=(mystery,True,100), K=(mystery,False,90), L=(science,True,80), M=(mystery,True,blank). Missing required fields request review.

Human Classifier Simulation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read all conditions

Three requirements must pass, not just a high preference score.



2. Trace J

Mystery, illustrated and exactly 100 pages: choose.



3. Trace K and L

K fails illustration; L fails genre: do not choose.



4. Trace M

Missing pages means review rather than assuming it is short.

CHANGE ONE THING

Supply M's pages as 101.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Read all conditions

What happens and why: Three requirements must pass, not just a high preference score.

Step: 2. Trace J

What happens and why: Mystery, illustrated and exactly 100 pages: choose.

Step: 3. Trace K and L

What happens and why: K fails illustration; L fails genre: do not choose.

Step: 4. Trace M

What happens and why: Missing pages means review rather than assuming it is short.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

This human simulation lets students inspect every decision without needing a trained model. Equality matters at 100. Known failures give reasons for rejection; M lacks enough information for the page check. The order should not turn blank into zero or quietly ignore a required field.

What this example does not establish

This declared rule checks eligibility, not whether an eligible book will actually be enjoyed.

Change one thing in the pictured case

Change: Supply M's pages as 101.

Prediction: M is now rejected rather than awaiting page review.

Reason: The required field is known but fails the maximum-length condition.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: Recommend if adventure AND pages<150; otherwise Other.

Write the question to explain: Trace A=(adventure,120), B=(adventure,180), C=(science,90).

Reveal after predictions: A is Recommend. B fails length; C fails genre. Show both conditions.

Keep for an independent attempt: A human classifier uses sports topic means sports shelf. Classify a

sports book and an unreadable cover. State what evidence is missing in the second case.

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Check presence, cm unit, whole number and range 1 through 10 before comparing. At 1, 6 and 10 cm predict short, tall and tall. At 11 cm record no prediction, outside range; at blank record no prediction, missing height. Record Tower1 throughout. Acceptance of 10 cm does not verify its label. Exit: 9 cm passes the contract and predicts tall, but correctness still needs evidence. A blank must stop as no prediction.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start 0 to 7 minutes

Teacher action: Retrieve threshold equality and the difference between missing and zero. Introduce the goal of instructions another learner can execute.

Stage: 7 to 20 minutes

Teacher action: Read the Tower1 contract and model aloud. Trace 5 cm, 6 cm and blank through the diagram. Stop physically before comparing the rejected card.

Stage: 20 to 33 minutes

Teacher action: Compare valid boundaries and rejected units or fractions. Complete W1 and explain why acceptance is not proof of a correct label.

Stage: 33 to 45 minutes

Teacher action: Start W2 in pairs. One learner writes, the other follows only the written instructions.

Require an explicit stop after a failed check.

Stage: Session 2 start 0 to 8 minutes

Teacher action: Retrieve the output record and both paths. Ask whether no prediction is a tower size.

Stage: 8 to 23 minutes

Teacher action: Use the guided task and W3 to test partners' procedures at endpoints, threshold and missing input. Record and repair ambiguity.

Stage: 23 to 38 minutes

Teacher action: Collect W4 independently before support. Look for transfer to the new range, threshold, labels and version instead of copied Tower1 constants.

Stage: 38 to 45 minutes

Teacher action: Use W5 and the exit to separate contract checks from correctness evidence. Preview final tests kept apart from model selection.

Reduce help deliberately

Completion task: Write numbered instructions for Tower1, including input checks, the stopping path after a failed check, the threshold rule and the record to keep. Your partner must be able to use them without supplying an unstated rule.

Teacher check: Check present height and cm unit, then whole number in 1 through 10. On failure, record original input, Tower1, no prediction and the reason; stop that record. Otherwise predict tall if height is at least 6, short otherwise. Record original input, Tower1 and prediction. Equivalent clear branches are acceptable.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Constructing a classifier. Proposed explanation: "Any numeric input must receive a category". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Check the stated unit, whole-number condition and supported range before

comparing the threshold.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Why are J and M initially treated differently even though both are illustrated mysteries?

Reasoned teacher answer

J has a valid including equality page count, 100. M's pages are unknown, so eligibility cannot be established.

Changed-case reasoning: M is now rejected rather than awaiting page review. The required field is known but fails the maximum-length condition.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Reveal six labeled creature cards, hide labels on four new cards, record each learner's rule and predictions, then compare disagreements.

What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Any numeric input must receive a category: Check the stated unit, whole-number condition and supported range before comparing the threshold.

An accepted input proves a correct prediction: Ask separately whether the input satisfies the contract and whether the prediction agrees with evaluation evidence.

No prediction means a third category of tower: It is a system outcome indicating that no short or tall category was assigned.

Writing a rejection is enough even if processing continues: Follow the failed path to an explicit stop so the later rule cannot overwrite the outcome.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner

restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

A is Recommend. B fails length; C fails genre. Show both conditions.

Independent question

A human classifier uses sports topic means sports shelf. Classify a sports book and an unreadable cover. State what evidence is missing in the second case.

Expected independent answer

The sports book goes to the sports shelf. The unreadable cover requires another source for the topic; do not infer it from appearance alone.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “Simulating a classifier removes the need for evidence.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

The first qualifies; the next two do not. The missing-genre case should request the genre or return uncertain under a documented policy. It should not be silently counted as a confirmed non-science book when reporting labelled evaluation results.

Explanations for the additional workbook examples

W1 Write an input contract

Write the complete Tower1 input contract. Give one accepted example and three different rejected examples, each with its reason. Explain why an accepted input does not have to be one of the examples used to fit the threshold.

Require a present height, unit cm, a whole number and range 1 through 10 inclusive. For example, 7 cm is accepted; blank is missing, 60 mm has the wrong unit and 11 cm is outside the range. Fractional values such as 4.5 cm also fail. The contract defines accepted inputs; it does not list only the fitting records or establish prediction correctness.

W2 Construct both outcome paths

Write numbered instructions for Tower1, including input checks, the stopping path after a failed check, the threshold rule and the record to keep. Your partner must be able to use them without supplying an unstated rule.

Check present height and cm unit, then whole number in 1 through 10. On failure, record original input, Tower1, no prediction and the reason; stop that record. Otherwise predict tall if height is at least 6, short otherwise. Record original input, Tower1 and prediction. Equivalent clear branches are acceptable.

W3 Trace boundaries and a broken path

Trace 1 cm, 5 cm, 6 cm, 10 cm and 11 cm using Tower1. A faulty procedure records no prediction for a blank, then continues and treats the blank as zero to report short. Identify the missing instruction and the proper final outcome.

The first four are accepted and predict short, short, tall and tall. At 11 cm reject for being outside the range. The faulty path lacks stopping after failure; a blank must end as no prediction with reason missing height. It must not be converted to zero or reach the threshold rule.

W4 Build an independent ribbon classifier

Design Ribbon1 for ribbon lengths recorded as whole centimetres from 1 through 12 inclusive, with unit cm required. Predict long at 8 cm or more and short otherwise. Write complete instructions and trace 7 cm, 8 cm, a blank length and 13 cm. Include reasons for no prediction and the model version in each record.

Check present length, cm unit, whole number and range 1 through 12; otherwise record no prediction with reason and stop. For accepted inputs, use at least 8 for long, otherwise short. Ribbon1 at 7 cm predicts short; at 8 cm predicts long; blank gives no prediction because length is missing; 13 cm gives no prediction because outside range. Each record includes original input, Ribbon1 and outcome, plus reason when rejected.

W5 Explain what your procedure establishes

Ribbon1 accepts 12 cm and predicts long. No comparison label has been supplied for that record. State what has been checked and what remains unknown. Explain why changing a blank to 1 cm just to produce a category would be a mistake.

The value satisfies the input contract and the stated rule predicts long. Agreement with an appropriate comparison label has not been checked. Replacing a blank with 1 invents a measurement, conceals missing information and violates the stated rejection path. Ask for the measurement instead.

Assess independent transfer

Selected case: A written rule selects science books with fewer than 140 pages. Trace science/139, science/140, history/90 and missing-genre/90. Policy: missing required information means request it.

Selected case answer

Select science/139. Reject science/140 at the strict boundary and history/90 on genre. Request the missing genre rather than inventing it. Both conditions are required.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design

prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Transparent Book Recommender

Turn score thresholds into visible boundaries such as recommend / maybe / not now.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record. Retrieve this record next week rather than restarting the project.

Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 11: Training and Testing. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Use separate check, compare and record cards with arrows. Let partners move each input physically and read the branch aloud. Extension: propose a separate mm-to-cm conversion stage; state what extra cases would need checking, including a conversion that yields a fractional centimetre.

W4 is secure with both paths, range 1 through 12, threshold 8 including equality, explicit stops and Ribbon1 records. After support retry Flag1: whole cm from 1 through 9, unit cm, large at least 5 otherwise small. At 4, 5, blank and 10 cm expect small, large, no prediction for missing and no prediction for range.

Week 15 keeps final evaluation examples unseen during fitting and explains why repeated changes based on their labels weaken the independence of that test.

Plan the delayed return

Suggested return points in this level: ai-2/11, ai-2/13, ai-2/16. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier

independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Build a transparent filtering stage for the recommender, storing each failed or unresolved requirement in its explanation.